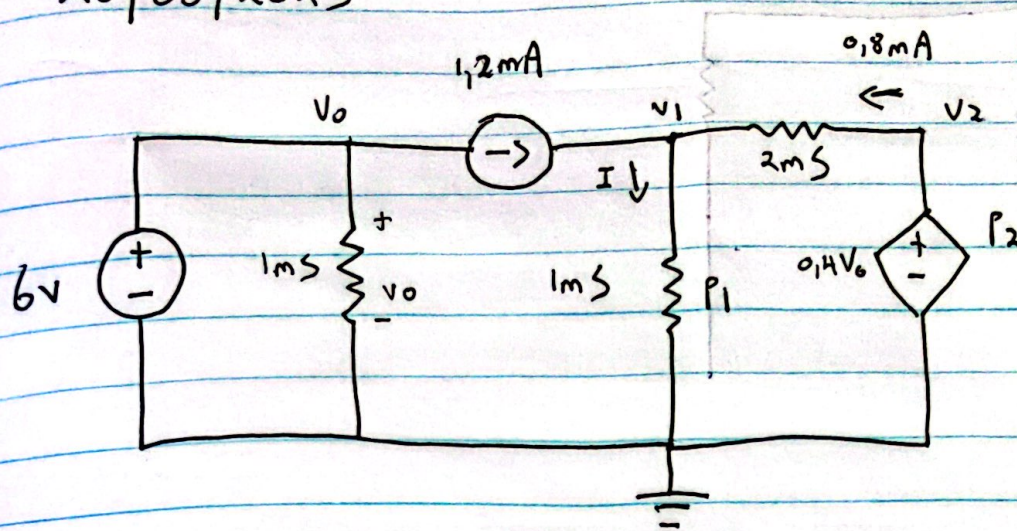


26/08/2023



$$S = \frac{1}{R} \quad R = \frac{1}{1 \times 10^{-3}} = 1000 \, \Omega$$

(vi) KCL $\Rightarrow$ $i_{in} = i_{out}$

$$1,2 \times 10^{-3} + 0,8 \times 10^{-3} = I$$

$$0,002 = I$$

$$P_1 = I^2 R$$

$$= (0,002)^2 (1000)$$

$$= 4 \, \text{mW}$$

P_2 $v_0 = 6 \, \text{V}$ (nodal)

$$v = 0,4 \times 6$$

$$= 2,4 \, \text{V}$$

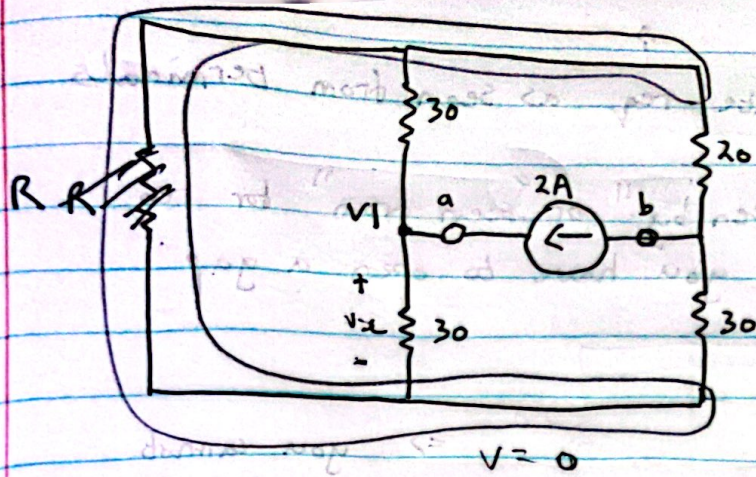
$$P_2 = \sum [v_2]$$

(entering from - side)

$$= - [2,4 \times 0,8 \times 10^{-3}]$$

$$= -1,92 \times 10^{-3}$$

$$= -1,92 \, \text{mW}$$



calculate something
"with respect to" a,b

a b
+ -

a) $R = \infty$
 $\Rightarrow$ remove resistor (disconnect)

find V_x

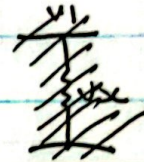
if you are able to
use current division.

$\Rightarrow$ replace it with nodal
analysis

nodal V_1

$$\frac{V_1}{60} + \frac{V_1}{30} - 2 = 0$$

$$V_1 = 54,55$$



$$V_x = \pm [I R]$$

$$= + \left[\frac{54,55}{60} \times 30 \right]$$

$$= 27,275 \text{ V}$$

b) $R = 0$ (close resistor with line)

find V_x

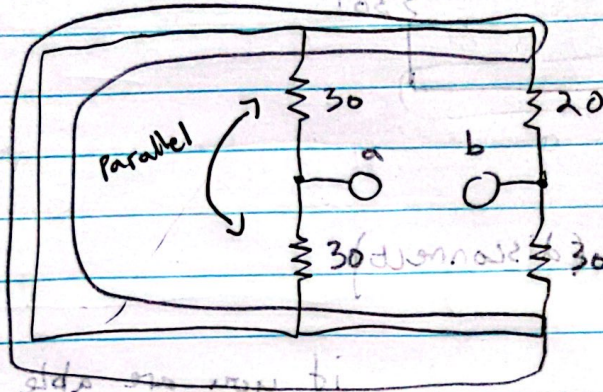
$$V_1 = V_x$$

$$\frac{V_x}{30} + \frac{V_x}{30} - 2 = 0 \quad V_x = 30 \text{ V}$$

equivalent

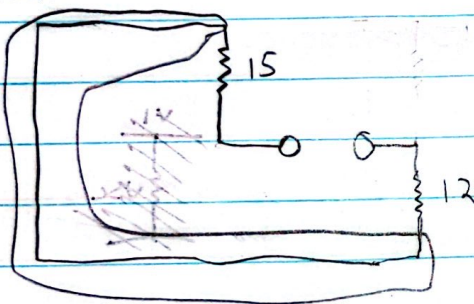
c) $R = 0$ calculate r_{eq} as seen from terminals

$\Rightarrow$ "seen by" or "seen from" for r_{eq}
 $\Rightarrow$ you have to open a gap



$\Rightarrow$ you cannot calculate r_{eq} if you don't identify the nodes

Kill one side if parallel



$\Rightarrow$ Parallel?

If 2 resistors branches share 2 or more nodes, they are parallel

$$V_1 = 24V, V_2 = 1V$$

Share 1 node

$$r_{eq} \rightarrow 15 + 12 = 27 \Omega$$

$$[R] \begin{bmatrix} I \\ I \end{bmatrix} = \begin{bmatrix} V \\ V \end{bmatrix}$$

$$\begin{bmatrix} 27 & 0 \\ 0 & 27 \end{bmatrix} \begin{bmatrix} I \\ I \end{bmatrix} = \begin{bmatrix} 24 \\ 1 \end{bmatrix}$$

$$V_1 = 24V, V_2 = 1V$$

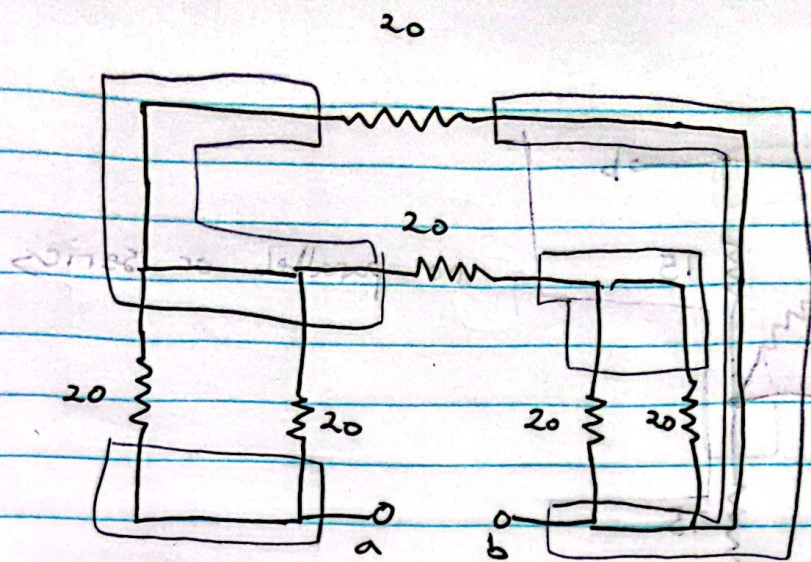
(close resistor with wire) $R = 0$

first

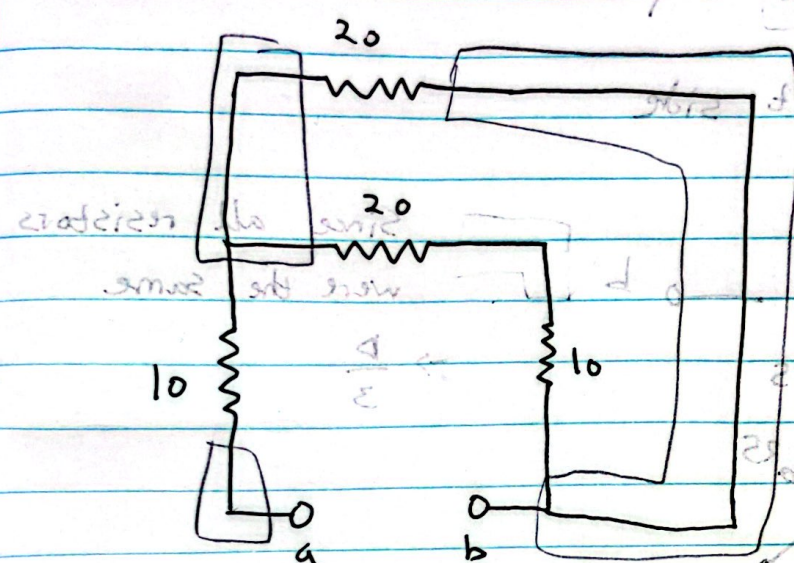
$$V_1 = 24V$$

$$V_2 = 1V$$

$$0 = \frac{24}{27} - \frac{1}{27} = \frac{23}{27}$$

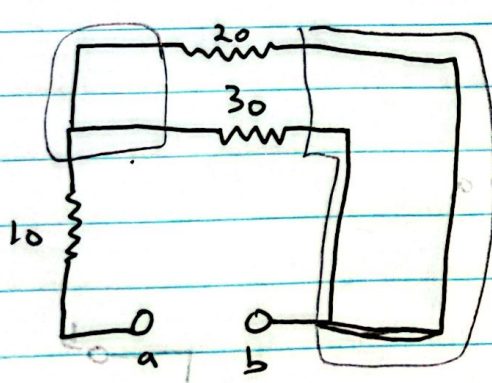


$$\left(\frac{1}{20} + \frac{1}{20} \right)^{-1}$$

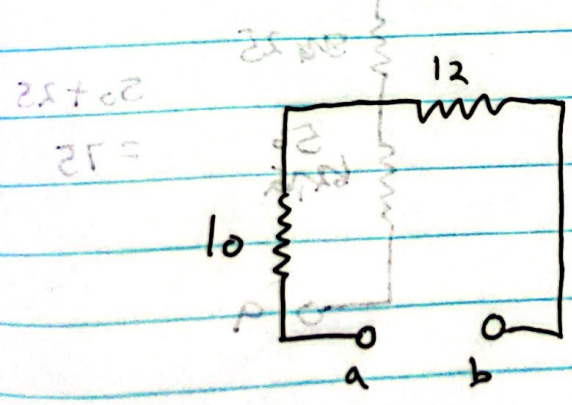


$10 + 10 = 20$

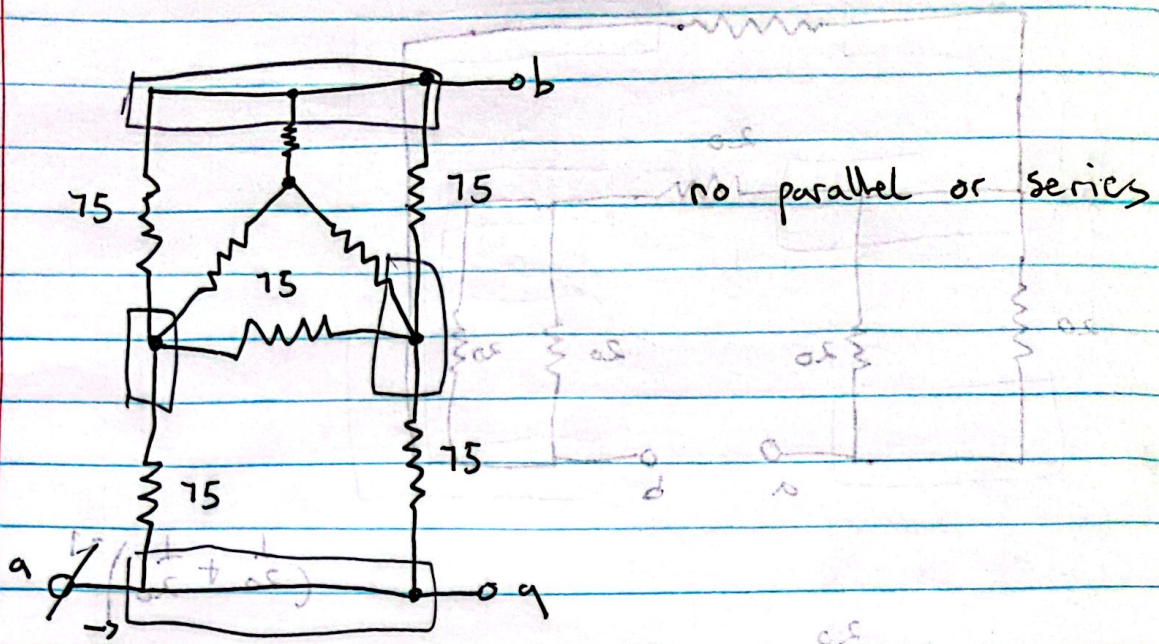
$10 + 20 = 30$



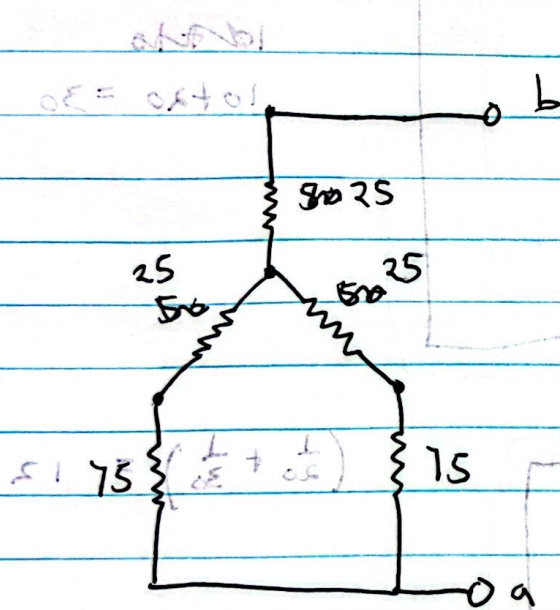
$$\left(\frac{1}{20} + \frac{1}{30} \right)^{-1} = 12$$



$10 + 12 = 22 \Omega$



move a to right side



Since all resistors were the same
 $\rightarrow \frac{D}{3}$

